Macroeconomics III / Assignment 2

Costs of businesscycles

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1 Business cycle empirics and costs

a) HP-filtering and business cycles

Using the python package `statsmodels.tsa.filters.hp_filter` i implement the HP filter i.e. I decompose each of the 3 dataseries for both countries by defining$y_t = \tau_t + c_t$, where the trend component τ_t solves

$$\min_{\{\tau_t\}_{t=1}^{T}} \left\{ \sum_{t=1}^{T} (y_t - \tau_t)^2 + \lambda \sum_{t=2}^{T-1} \left[(\tau_{t+1} - \tau_t) - (\tau_t - \tau_{t-1}) \right]^2 \right\},$$

where λ indicate the level of regularization i.e. the degree that volatility of the trend-component is "discouraged".

Figure 1 plots the resulting trend deviations $\{c_t^{\lambda=1600}\}_{t=1995Q1}^{2026Q2}$.

b) Comparing volatility

As shown in Table 1, investment is substantially more volatile than output in both countries, with relative volatilities of 4.26 in the US and 2.23 in Portugal. Consumption is less volatile than output in Portugal ($\sigma_c/\sigma_y = 0.83$), whereas the US exhibits slightly higher consumption volatility ($\sigma_c/\sigma_y = 1.05$), contrary to the usual empirical regularity.

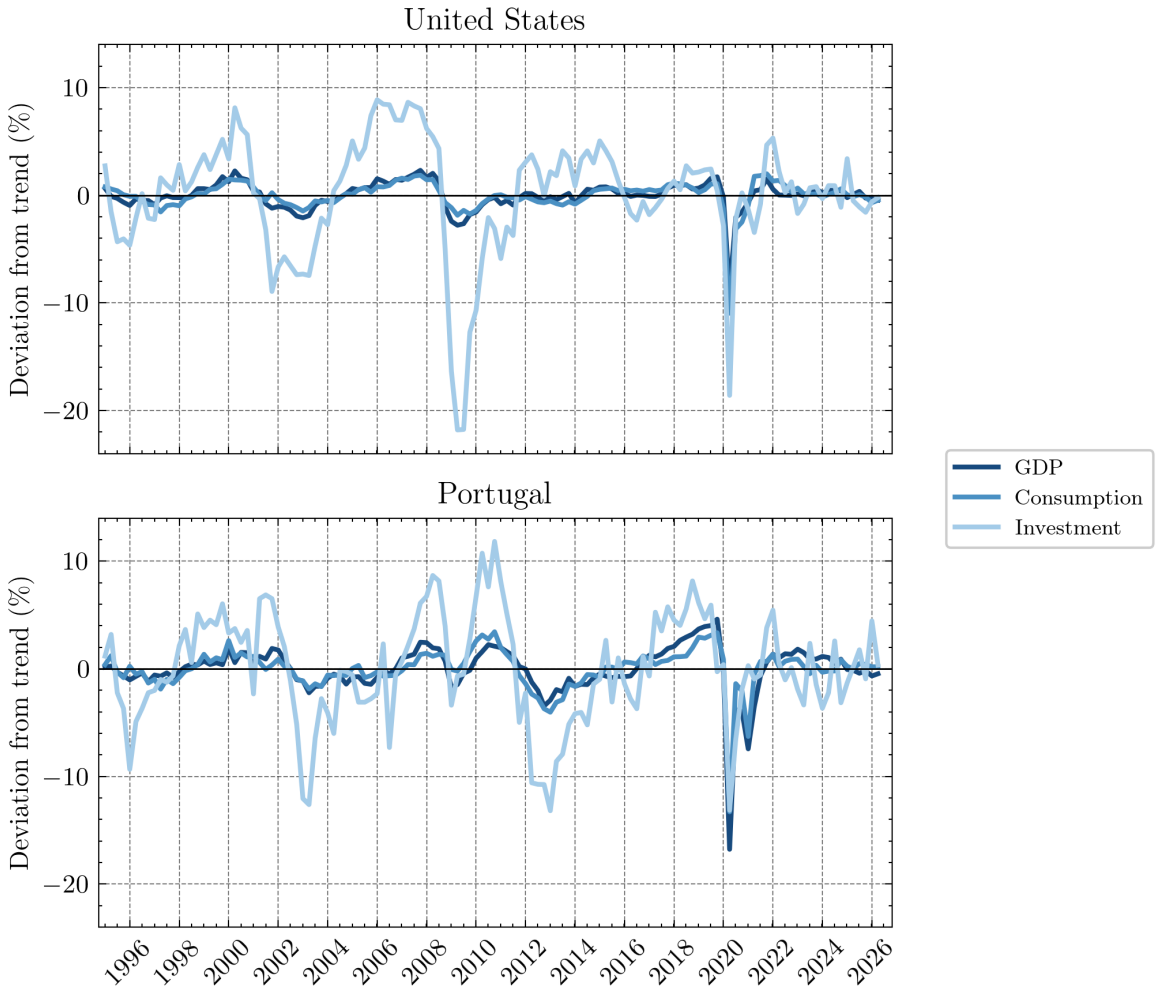
Overall, Portugal experiences greater output and consumption volatility, while investment is slightly more volatile in the US. Thus, both countries exhibit pronounced investment fluctuations, but differ considerably in their relative business cycle volatilities.

Country	σ_y	σ_c	σ_i	σ_c/σ_y	σ_i/σ_y
United States	1.28	1.35	5.45	1.05	4.26
Portugal	2.27	1.89	5.06	0.83	2.23

Tabel 1: Business Cycle Volatility: United States and Portugal

Note: Standard deviations are reported in percent, while ratios are unitless. Cyclical components are obtained by applying the Hodrick-Prescott filter ($\lambda = 1600$) to logarithmic quarterly series from 1995Q1 to 2026Q2.

Sources: FRED (GDPC1, PCECC96, GPDIC1) for the United States; Eurostat (namq_10_gdp) for Portugal. Own calculations.



Figur 1: Trend deviations for selected variables in USA and Portugal

Note: The graphs show the cyclical component resulting from a hp-filtering applied with $\lambda = 1600$ which is standard for quaterly data.

Sources: FRED (GDPC1, PCECC96, GPDIC1) for the United States; Eurostat (namq_10_gdp) for Portugal. Own calculations.

c) Cost of representative risk

First we note that identically distributed shocks gives $\mathbb{E}\left[u\left(\left(1+\epsilon_t\right)\bar{c}\right)\right] = \mathbb{E}\left[u\left(\left(1+\epsilon_{t+k}\right)\bar{c}\right)\right]$, $k \geq 0$ implying

$$\begin{aligned}\sum_{t=0}^{\infty}\beta^t u\left(\left(1-\lambda_c\right)\bar{c}\right) &= \mathbb{E}\left[\sum_{t=0}^{\infty}\beta^t u\left(\left(1+\epsilon_t\right)\bar{c}\right)\right] \\ &= \sum_{t=0}^{\infty}\beta^t \mathbb{E}\left[u\left(\left(1+\epsilon_t\right)\bar{c}\right)\right] \iff \\ u\left(\left(1-\lambda_c\right)\bar{c}\right) &= \mathbb{E}\left[u\left(\left(1+\epsilon_t\right)\bar{c}\right)\right].\end{aligned}$$

Using the CRRA utility specification and isolating λ this yields

$$\begin{aligned}\mathbb{E}\left[\frac{\left[\left(1+\epsilon_t\right)\bar{c}\right]^{1-\gamma}-1}{1-\gamma}\right] &= \frac{\left[\left(1-\lambda\right)\bar{c}\right]^{1-\gamma}-1}{1-\gamma} \iff \\ \mathbb{E}\left[\left(1+\epsilon_t\right)^{1-\gamma}\right] &= \left(1-\lambda\right)^{1-\gamma} \iff \\ \lambda &= 1-\left(\mathbb{E}\left[\left(1+\epsilon_t\right)^{1-\gamma}\right]\right)^{\frac{1}{1-\gamma}}\end{aligned}$$

Using that ϵ_t is uniformely distributed and $\epsilon_t \in [-\epsilon, \epsilon]$ we have

$$\begin{aligned}\sigma_c &= \frac{1}{\sqrt{12}}\left[\left(1+\epsilon\right)-\left(1-\epsilon\right)\right] \iff \\ \epsilon &= \sigma_c\frac{\sqrt{12}}{2}\end{aligned}$$

and using the hint

$$\mathbb{E}\left[\left(1+\epsilon_t\right)^{1-\gamma}\right] = \frac{1}{2\epsilon}\frac{1}{2-\gamma}\left[\left(1+\epsilon\right)^{2-\gamma}-\left(1-\epsilon\right)^{2-\gamma}\right]$$

we obtain the final expression for λ as a function of parameters and the

standard deviation of consumption:[1]

$$\lambda_c = 1 - \left[\frac{1}{2\epsilon}\frac{1}{2-\gamma}\left[(1+\epsilon)^{2-\gamma} - (1-\epsilon)^{2-\gamma}\right]\right]^{\frac{1}{1-\gamma}}$$
$$= 1 - \left[\frac{1}{\sigma_c\sqrt{12}}\frac{1}{2-\gamma}\left[\left(1+\sigma_c\frac{\sqrt{12}}{2}\right)^{2-\gamma} - \left(1-\sigma_c\frac{\sqrt{12}}{2}\right)^{2-\gamma}\right]\right]^{\frac{1}{1-\gamma}}.$$

The subscript c simply indicates that this depends on the country (i.e. USA and Portugal).

Results are displayed in table 2 where the first column corresponds to $\gamma = 1.99$.

Country	$\gamma = 1.99$	$\gamma = 4$	Employment risk ($\gamma = 2$)
United States	0.0182	0.0365	1.1875
Portugal	0.0355	0.0714	1.1875

Tabel 2: Welfare Costs: United States and Portugal

Note: Costs are reported as the percentage of mean consumption an individual would give up to eliminate risk. The first two columns show business cycle costs. Employment risk assumes consumption of $1.02\bar{c}$ with 95% probability and $0.62\bar{c}$ with 5% probability, and is identical across countries.

Sources: FRED (PCECC96) for the United States; Eurostat (namq_10_gdp) for Portugal. Own calculations.

d) Cost of representative risk with higher risk aversion

Looking comparing the two first columns of table 2 shows how the business cycle costs has increased with γ.

The intuition is simple: larger γ implyies stronger diminishing marginal utility of consumption. Therefore in booms the agent with $\gamma = 4$ does not really increase utility and in downturns they heavily decrease utility compared to an agent with a lower $\gamma = 1.99$.

Note in addition to this is a simple utility model. There is for instance no

[1]The intepretation of λ is the share of consumption that the consumer is willing to forego in order to not experience consumption fluctuations compared to the scenario where the they experience uniformly distributed fluctuations with standard deviation σ_c.

long term effects of jobloss.[2] This is simply one example but one could come up with many (importance of consumption relative to peers, social status loss of unemployment, ,...). Therefore the exact numbers should *not* (in my opinion) be interpreted as meaningfull estimates of business cycles costs. Instead the exercise should be used to highlight the *increase* in business cycles costs when considering the *individual large fluctuations* as supposed to the *aggregate* relatively *smooth fluctuations* in consumption.

e) Cost of unemployment risk

Using the exact same logic as in c we obtain λ that makes the consumer indifferent between the two economies:

$$\lambda = 1 - \left[0.95 \left(1 + \frac{\sigma}{\sqrt{19}} \right)^{1-\gamma} + 0.05 \left(1 - \sqrt{19}\sigma \right)^{1-\gamma} \right]^{\frac{1}{1-\gamma}}.$$

Table 2 displays how the losses are way larger here. The lesson is important: when observing aggregate consumption fluctuations this does not nessesarily reflect the individual fluctuations in consumption since they may somewhat smooth out. i.e. when one person becomes unemployed another becomes employed and vice versa. So in aggregate σ_c is relatively low compared to the idiosyncratic $\sigma_{c,i}$.

Thus our first *representative* analysis (probably) understates the negative consequences of business cycles. [3]

[2] For current danish context i refer to the latest DØRS report chapter IV that tries to empirically estimate how uncertainty in employment affects stress levels and long term unemployment. They find that jobuncertainty is among the very worst for severe stress related illness and early retirement.

[3] Note that we assume some level of uninsurable risk that is somewhat independent between individuals such that consumption actually fluctuates more on an individual level than in aggregate.

2 Multiple choice question

a) Ramsey-Cass-Koopmans model

Consider a standard discrete-time Ramsey economy with fixed labour supply and no population or productivity growth. Let $f(k)$ denote output per person, δ the capital depreciation rate, and $\beta \in (0,1)$ the household discount factor.

The government permanently increases purchases per person by $\Delta g > 0$, financed by lump-sum taxes. These purchases provide no utility and do not affect production technology.

Comparing the two interior steady states, what happens to capital k and private consumption c per person?

A. k increases and c decreases
B. k decreases and c decreases
C. k stays the same and c decreases by exactly Δg
D. k stays the same and c stays the same
E. k stays the same and c decreases by less than Δg

Correct answer: C.

Saving one additional unit at date t reduces current consumption by one unit and provides $1 + f'(k_{t+1}) - \delta$ additional units of consumption next period. At an interior optimum, the marginal utility cost equals the discounted marginal utility benefit:

$$u'(c_t) = \beta u'(c_{t+1})\big[1 + f'(k_{t+1}) - \delta\big].$$

This is the Euler condition.

In a steady state, consumption and capital are constant. Cancelling $u'(c^*) > 0$ gives

$$1 = \beta\big[1 + f'(k^*) - \delta\big],$$

or

$$f'(k^*) = \frac{1}{\beta} - 1 + \delta.$$

Government purchases do not enter this condition, and lump-sum taxes do not distort the return to saving. Since technology, preferences, and depreciation are unchanged, steady-state capital k^* remains unchanged.

The aggregate resource constraint is

$$c_t + k_{t+1} = f(k_t) + (1 - \delta)k_t - g_t.$$

In a steady state, $k_{t+1} = k_t = k^*$, so

$$c^* = f(k^*) - \delta k^* - g.$$

With capital unchanged, output and replacement investment are unchanged. Therefore,

$$c^*_{\text{new}} - c^*_{\text{old}} = -\Delta g.$$

Thus, capital stays the same and private consumption falls by exactly Δg.

Why the other options are incorrect.

A and B incorrectly assume that higher government purchases change steady-state capital. In particular, B may reflect the mistaken belief that lump-sum taxes distort saving. D ignores the resources absorbed by government purchases.

E is tempting because households smooth consumption. However, smoothing does not create additional steady-state resources. With output and replacement investment unchanged, a consumption decrease smaller than Δg would violate the resource constraint.

b) OLG model

In a standard two-period OLG model, the government introduces a compulsory, fully funded pension. Each young worker contributes $p > 0$, earning the same return as voluntary saving. Workers can freely borrow and save, labour supply is fixed, and there are no other taxes or fees.

How does voluntary saving change?

A. It increases by p

B. It stays unchanged

C. It decreases by less than p

D. It decreases by exactly p

E. It decreases by more than p

Correct answer: D.

Let s denote voluntary saving, w wage income when young, and R the gross return on both voluntary saving and pension contributions. Consumption when young and old is

$$c_y = w - p - s, \qquad c_o = R(s + p).$$

Define total saving as $S = s + p$. The budget constraints become

$$c_y = w - S, \qquad c_o = RS.$$

These are the same consumption possibilities as before the pension was introduced. Preferences are also unchanged, so the worker's desired total saving S^* remains unchanged.

Before the pension, voluntary saving equals S^*. After its introduction, voluntary saving equals $S^* - p$. Therefore,

$$s^*_{\text{new}} - s^*_{\text{old}} = -p.$$

Voluntary saving falls by exactly p, while total saving, capital, and consumption in both periods remain unchanged. Free borrowing ensures that this adjustment is feasible even if voluntary saving becomes negative.

Why the other options are incorrect.

B is tempting because it confuses total saving with voluntary saving. If voluntary saving stayed unchanged, the pension contribution would increase total saving by p.

A and C also imply an increase in total saving, while E implies a decrease. Each contradicts the worker's unchanged desired total saving.

c) Real Business Cycle model

A worker chooses hours $h \in (0,1)$ before knowing their wage $w > 0$. Consumption is $c = wh$, and expected utility is

$$E[\log c + \log(1-h)].$$

Suppose the wage becomes riskier through a nontrivial mean-preserving spread. All expectations are finite.

What happens to the worker's optimal hours and expected utility?

A. Hours increase and expected utility falls.
B. Hours decrease and expected utility falls.
C. Hours stay the same and expected utility stays the same.
D. Hours stay the same and expected utility falls.
E. Hours increase and expected utility rises.

Correct answer: D. Substituting the budget constraint gives

$$E[\log w] + \log h + \log(1-h).$$

Wage risk only affects the first term, so it does not change optimal hours. This rules out A, B and E. Since the logarithm is strictly concave, the mean-preserving spread lowers expected utility, ruling out C.